James Seidel

(512) 757-2980 | jameseseidel@icloud.com | linkedin.com/in/jamesseidel

EDUCATION

University of Texas at Austin, Austin, TX **May 2029**
Bachelor of Science, Mechanical Engineering
Overall GPA: 4.00/4.00
Relevant Coursework: Thermodynamics, Statics, Diff. Eqns. w/ Linear Algebra, Intro to Engr. Design and Graphics

ENGINEERING EXPERIENCE

Engineering Intern | *Southwest River Engineering, 06/2026 – 07/2026*
- Executed hydraulic analysis using HEC-RAS to simulate flood events, verify safety of construction plans, and secure regulatory floodplain permits.
- Developed 3D surface models and detailed plan sets using **Civil 3D**, transforming raw survey data into conceptual designs for projects like pond dredging, habitat restoration, and drainage channel design.
- Conducted daily site inspections and maintained technical logs to document construction progress and ensure contractor compliance with engineering specifications.

Systems Subteam Member | *Texas Flight, 09/2025 – Present*
- Design and fabricate banner deployment and cargo storage mechanisms using **SolidWorks** and 3D printing for competition aircraft.
- Optimize plane components to meet competition objectives, including secure cargo storage, weight limits, and in-flight banner deployment.
- Collaborate with a 20+ member team in weekly meetings to develop and iteratively test aircraft components.

Mechanical Lead | *Robotathon (UT IEEE RAS), 09/2025 – 11/2025*
- Led the mechanical design and fabrication of a projectile launching mechanism for a semi-autonomous robot using **Onshape** and 3D printing (awarded **Best Design** of the competition).
- Coordinated with a 10-person team to integrate mechanical systems with autonomous maze navigation and color-sensing subsystems.

ENGINEERING PROJECTS

Drone Interceptor | *Personal Project, 06/2026 – Present*
- Design a low-cost, 3D-printable airframe in **SolidWorks**, optimizing external geometry and aerodynamic profile to maximize flight efficiency and endurance.
- Analyzed existing drone designs to select optimal propulsion sizing, weight distribution, and modular 3D-printable airframe components.

Odor-Reducing Laundry Hamper | *Design Lead, 10/2025 – Present*
- Spearhead the structural design and manufacturing of a specialized laundry hamper engineered to actively reduce odors through airflow management and clothing storage.
- Create detailed part and assembly models in **Onshape**, balancing structural integrity and manufacturability.
- Test prototype performance by measuring impacts of airflow velocity and humidity levels on odor mitigation.

QUALIFICATIONS

- **CAD:** SolidWorks, Onshape, AutoCAD, AutoCAD Civil 3D
- **Manufacturing:** 3D Printing, Rapid Prototyping, Laser-cutting, Soldering
- **Honors:** University Honors (Fall 2025, Spring 2026), National Merit Scholar Finalist (2024)